

CS201 Midterm 1 Part A

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I. QUESTION (ALGORITHM ANALYSIS)

```
1 int algorithm1(int N){  
2   if (N == 0)  
3     return 0;  
4   int sum = 0;  
5   for (int i = N; i > 0; i--)  
6     for (int j = 0; j < N; j++)  
7       sum++;  
8   for (int i = 0; i < 4; i++)  
9     for (int j = 0; j < 4; j++)  
10      sum += algorithm1(N / 4);  
11   return sum;  
12 }
```

What is the time complexity of algorithm1? Solve the recurrence equation without the master theorem.

II. QUESTION (ALGORITHM ANALYSIS)

```
1 int algorithm2(int N){  
2   int i, j, k, sum = 0;  
3   for (i = 1; i <= N; i++)  
4     sum++;  
5   for (i = 1; i <= N; i++)  
6     for (j = 1; j <= 2 * i; j++)  
7       for (k = 1; k <= N; k++)  
8         sum++;  
9   for (i = 0; i <= N; i++)  
10    for (j = i + 1; j <= N - i; j++)  
11      sum++;  
12   return sum;  
13 }
```

What does the function algorithm2 return in terms of N ? Assume that N is even.